OPEN ITEMS CHECKLIST — part 7 (19 October 2026)

Changes since part 6 (17 October 2026).

Programming-Tools conditional compilation (stage 6d)

- ☑ **`[IF]`, `[ELSE]`, `[THEN]` added to the ROM** (immediate; headers `H_BRIF`, `H_BRELSE`, `H_BRTHEN`; code `BRIFW` / `BRELSEW` / `BRTHENW`, helper `BRCMP`). One shared skip routine scans with `PARSE-NAME`, counts nested `[IF]`s on the return stack, matches the three names exactly (case matters, as for every name here; a first version folded case, which was inconsistent because the interpreter never finds `[if]` anyway) and calls `REFILL` when the line runs out; it stops at the end of an `EVALUATE` string. Cost: 144 bytes of code and 31 of headers.
- ☑ **Assembler unit test `TSTBRCOND`** (group 17, `TSTSELECTOR=17`), table driven: skip to `[ELSE]`, nested `[IF]`s, lower and mixed case names ignored, `[THEN]` only, no terminator, true flag, `[ELSE]` entered directly, look-alike names.
- ☑ **Checked off target.** `TSTBRCOND` and the three words were run on a small 6809 emulator (not MAME). It found a real defect in the first version of the test (`LDD` destroyed the mode held in `A`), now fixed. 3000 random inputs matched a model of the standard algorithm, with the data stack and return stack unchanged.
- ☑ **ROLL defect found by test 30 in MAME and fixed.** `ROLL` read u with `PULU D` and branched with `BEQ`; `PULU` does not set the condition codes, so with u >= 1 and Z set it did nothing. The unit test passed by luck (the `STU` before the call always clears Z). Now `LDD ,U++` (same size) sets the flags and `TSTROLL` leaves Z set before the call. Checked on the emulator: the old code fails the new test, the new code passes it and 500 random cases. Same pattern elsewhere: a scan of every branch whose previous instruction does not set the flags found only `FINDW` (`TFR A,B` then `BEQ FMATCH`, where the flag comes from the `LEAX` before it, so that branch never fires; unreachable for a real name, since an empty name is rejected earlier) - left as is.
- ☐ **MAME confirmation** of `TSTBRCOND` (group 17) and of the new ANS file `31_tools_conditionals.tests.fs`. The last two tests there span several lines, so the runner waits for a reply timeout on each interior line.

- ☐ **Remove `00a_tool_ext_conditionals.fs`** from the harness list in `ans_test_runner.py` (the `for fname in (...)` line near the end of `main`), from `run all tests.sh` if it is loaded there, and from the README load order (the README in this delivery is already updated). The runner's static stripping of ttester's `HAS-FLOATING` blocks can stay: it still avoids a reply timeout for every line the target would skip.

ROM layout

- ☑ **`BASECODE` and `BASEDICT` lowered** to `$DBF0` and `$D322` (were `$DC44` and `$D395`). Polled `BASECODE` is 8803 bytes (`$DBF0` - `$FE52`), interrupt-driven 9122 (`$DBF0` - `$FF91`), `BASEDICT` 2254 (`$D322` - `$DBEF`). Padding below `INITCODE`: 337 bytes polled, **18 bytes interrupt-driven**.
- ☑ All 19 unit-test selectors (0 to 18) assemble in all four builds.
- ☐ **Interrupt-driven headroom is 18 bytes.** The next addition to the base code needs another downward shift (the unused ROM below `BASEDICT` is now 4642 bytes, shared with the unit-test code) or a saving.

New ANS tests

- ☑ **`30_coreext_misc.tests.fs`** for the 25 words that had no ANS test and are testable (see the README). Values for `WITHIN` and `U>` were checked against a model of 16-bit arithmetic. MAME run by hand: `ROLL` failed (real bug, fixed above); `TIB 0106 =` failed only because the user's abbreviated test framework had not set the base to HEX (the test is unchanged and correct). [x] **MAME re-run: all 113 tests in the file pass** with the ROLL fix and the HEX base. Assumptions to watch: `#TIB @ SOURCE NIP =`, `1 SNAME NIP -> 0`.
- ☐ Words with no ANS test, by nature: `KEY`, `KEY?`, `EXPECT`, `QUERY`, `SPAN`, `QUIT`, `.S`, `WORDS`, `DUMP`, `?` (interactive or print only).

Documentation

- ☑ `ClaudeForth` docx and PDF (19 October 2026, second issue): three new glossary and index entries (245 words), memory map table (new sizes, the unused ROM row now `$C100-$D321`, 4642 bytes), ROM size text, Section 4.3 paragraph on conditional compilation, Section 5 items, test-runner text without `00a`, regenerated source appendix (stage 6d). Memory-map SVG and PDF regenerated.